CSD Exam 3 (90 marks, Time= 90 min)

Instructions

- *This is a closed book exam. One single sheet of DB schema and formulae is permitted. It needs to be submitted at the end of the exam*
- *Any use of unfair means will lead to a disqualification from the exam.*
- *Use of a calculator is not necessary but permitted.*
- *Clearly state any assumptions you need to make to solve the problem*

Q1. [30 marks] Citizen 360 system

The government is building a Citizen 360 system to pro-actively reach out to citizens and inform them about various welfare schemes, and opportunities, assess if a scheme has been successful, measure citizen satisfaction, pain points, etc.

For this question, the relevant information is stored in the following. Assume that the information is updated once every year.

Aadhar DB: Aadhar ID Number, Name, DOB, Father's Name, Address, Phone number [1 billion Records]

Tax DB: Date, Name, Aadhar ID Number, PAN, Yearly Income, Tax Paid Amount, Bank Account Numbers [1 billion Records]

Education_Enrol_DB: Date, School/College Name, Student Name, Student Aadhar ID Number, Degree Registered For [1 billion Records]

Education_Pass_DB: Date, School/College Name, Student Name, Student Aadhar ID Number, Degree Obtained [1 billion Records]

Employment DB: Date, Employer, Employee Name, Employee Aadhar ID Number, Address, Phone Number, Salary [1 billion Records]

Note that many fields may be empty or even become obsolete.

The Government has released a *Skill development scholarship* program for all students who satisfy following criterion:

a. Have passed 12th class (Senior Secondary) within last 5 years

b. Currently unemployed and not enrolled in a higher education institute

c. Yearly family income is less than 2 lakhs.

Implement the following services in an efficient manner:

[10 marks] CHECK_ELIGIBILITY (App_Data**):** Verifies if a candidate is eligible (in this case for Skill development Scholarship) based on the data present in the application form: Name, Aadhar ID Number, DOB, Father's Name, Mother's Name, Address, Phone number, Family Income, School for 12th, Marks Obtained, Employment Status.

Analyze the running time of the above API in terms of the number of operations assuming each DB has N=1 billion records. Optimize the system to minimize this running time.

Note: Clearly state how entities will be resolved.

[20 marks] **NOTIFY_SCHEME**: Sends a notification message as SMS to all eligible candidates in the DB providing information about a given scholarship, in this case, *Skill development scholarship*. Analyze the running time of the above API in terms of the number of operations assuming each DB has N=1 billion records. Optimize the system to minimize this running time.

Note: Clearly state how entities will be resolved.

Q2 [30 marks] DB Questions

2a. [7 marks] Mark the following statements in regards to SQL/NoSQL DBs as **True or False.**

i. NoSQL DBs allows dynamic schemas where columns/attributes can be added on the fly.

ii. SQL DBs are quite powerful in defining and manipulating the data.

iii. NoSQL DBs are less cost-effective for big data use cases than SQL DBs.

iv. NoSQL DBs are usually the best fit for financial applications where data integrity is a must.

v. SQL DBs allow rapid development in scenarios where your application and data structures are frequently updated.

vi. Dynamo always writes data to the first 'N' nodes in the *preference list*.

vii. LSM trees have smaller write amplification as compared to B+ trees during appends.

2b. [4 marks] Suppose that the **TrueCaller** app needs to implement a service to flag Spam calls. What would be the most efficient way to do this? Describe briefly (less than 100 words).

2c. [3 marks] An online fintech service that supports end-to-end digital loan origination which provides support for automated application and decisioning for commercial loans. They provide a firm-wide view of different loan applications in various stages and across various geographies without compromising on consistency of data. This requires providing a single view to the system that integrate different views of the data.

Between DynamoDB and MongoDB which one would you use in the following use case? Justify your answer.

2d. [2 marks] Amazon Dynamo is called a zero-hop DHT because:

i) A collision-free hash function is used.

ii) Consistent hashing allows the system to grow.

iii) Clients know the complete set of servers and can directly contact the server that holds the data they need.

iv) Each server node maintains a finger table that identifies other nodes.

2e. [2 marks] Consider following case in a DynamoDB deployment comprising 7 servers and a replication factor of 3. During a write operation if a replica server X is temporarily down or unreachable, the operation is sent to an alternate server Y. The updates value is sent to X whenever X comes up. The above functionality is provided by which mechanism/protocol in DynamoDB?

i. Merkle Tree

ii. Sloppy Quorum

iii. Hinted handoff

iv. Gossip protocol

2f. [2 marks] What are the benefits of Vnodes in Dynamo? Select all that apply.

i. Vnodes help spread the load more evenly across the physical nodes on the cluster. This speeds up the rebalancing process after adding or removing nodes.

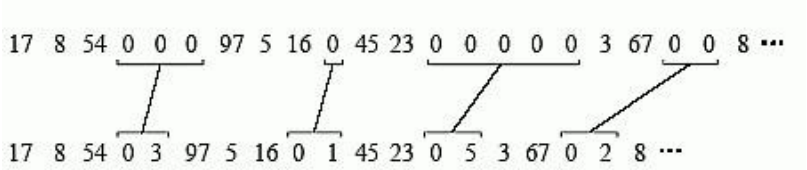
ii. Vnodes enable the use of virtual machines to help Dynamo scale better.

iii. Vnodes make it easier to maintain a cluster containing heterogeneous machines.

iv. Use of Vnodes decreases the probability of hotspots, as Vnodes help assign smaller ranges to each physical node.

2g. [2 marks] In C-Store, when can you perform queries directly on the compressed data? Justify why.

2h. [2 marks] With reference to C-store, what kind of encoding scheme operates in the following way.



2i. [2 marks] What is the importance of a WAL (Write Ahead Log) in Log Structured Merge (LSM) Trees?

2j. [4 marks] In Meta (Facebook), a C-Store is used to store the information regarding user's pages. With reference to WOS and ROS, please explain the loading and querying process. Why do we need to move tuples between WOS and ROS?

Q3 QUEUING THEORY QUESTIONS

3a. [5 marks] Tech Café is free 20% of time (i.e. no customer present). Assuming a Poisson arrival process and exponentially distributed service times, what is the probability that a random observer will find 2 customers?

3b. [6 marks] In the city of Raipur, a new city bus service is introduced, which comes to GEC bus stop quite irregularly. The average time between bus arrivals going towards the Raipur city is 10 minutes, but the std. deviation is 30 minutes. What is the average amount of time a student has to wait for the bus at the GEC bus stop to go to Raipur?

3c. [10 marks] Google Colab gets 10 requests per second for users trying to connect their Colab Notebooks. If an average user spends 30 minutes actively doing his work, what is the resource requirement (R)? If Google wants to give 1 vCPU immediately to 90% of their users, how many servers (with 200 vCPU each) does Google need to run the Colab service?

3d. [9 marks] Indigo's check-in counter at Delhi International airport has a single queue. There are 4 Airline agents and the customer is sent to the next available agent. The arrival rate is 2 people/min according to Poisson process; average service time is 1 min (exponential distribution) per ticketing agent. State if each of the statements is True or False with brief justification.

Note: You won't get any credit if the justification is missing or incorrect

i. The check-in counter is empty roughly 50% of the time

ii. Suppose the passengers go to an automated Check-in Kiosk if all Airline agents are busy. The probability of going to automated Kiosk is higher than that of having to wait if there was no Automated Kiosk.

iii. The airline can reduce number of ticketing agents to 3, without much difference to response time